



NGS Genotyping Pipeline Documentation

This contains all the documentation for the NGS Genotyping Pipeline, NGSG Retype, and NGSG Reference, as well as how to get started using the NGSG Explorer web service v2.04.000.



Software installation

To install the pipeline and additional tools (Retype and Reference), follow these steps:

1. Clone the repository from GitHub. \$ git clone <https://github.com/cameron-jack/NGSgeno.git>
2. Create a new virtual environment. \$ python -m venv nsgsgeno_env
3. Activate the virtual environment. \$ source nsgsgeno_env/bin/activate
4. Install Python v3.12 and the required dependencies using pip. \$ python -m pip install .
5. Create desktop shortcuts to the run scripts: *Run_NGSgeno.bat*, *run_reference.bat*, and *run_retype.bat*
6. Double click your shortcut to run the application.

Using the pipeline

New users should read:

- [the glossary](#) to familiarise themselves with the terms used in this pipeline
- [the start page](#) that they will see when they first start the application
- [the new experiment documentation](#) to understand the standard features of the interface.

The NGS Genotyping pipeline is composed of several stages:

1. [Load samples, consumables, amplicon plates](#)
2. [Hamilton Nimbus plate combining \(96 to 384 well\)](#)
3. [Primer selection and PCR step 1](#)
4. [Indexing and PCR step 2](#)
5. [Miseq sequencing](#)
6. [Allele matching](#)
7. [Reporting](#)

Additional software tools

There are three additional stand-alone tools that have been integrated with v2.04.000 of the pipeline:

- [NGSG Version Select](#) - checks for new pipeline versions and allows the user to choose which one they want to be active
- [NGSG Retype](#) - for finding failed assay runs and building a new custom manifest (384 well) to rerun (retype) these assays
- [NGSG Reference](#) - which take an Excel-based reference file and converts it to FASTA, while also looking for forbidden characters and formatting issues

Technical Details

This section provides detailed information about technical aspects of the pipeline where various choices will impact the outcomes.

- [Info viewer panels](#)
- [Activity log](#)
- [Input and output file formats](#)
- [Sequence matching](#)

Authorship

This project was developed by the staff of the ANU Bioinformatics Consultancy (ABC) and the Genotyping team of the Biomolecular Resource Facility (BRF), at The John Curtin School of Medical Research, The Australian National University.

License

This project is licensed under the MIT License. See the LICENSE file for more details.



NGS Genotyping Pipeline Documentation: Glossary

Here is an alphabetical list of terms used in the NGS Genotyping Pipeline. It is by no means exhaustive, but aims to provide a useful reference for new users.

Jump to: [A-D](#) [E-H](#) [I-L](#) [M-P](#) [Q-T](#) [U-Z](#)

A-D

Amplicon - a piece of DNA that is the source and/or product of amplification or replication events (e.g. PCR products)

Alignment - the process of arranging sequences to identify regions of similarity, aka *matching*

Allele - a variant form of a gene that occupies a specific position on a chromosome. An allele may take several forms and require multiple primers for detection, and many different target sequences for matching

Adapter clipping - the process of removing adapter sequences from the ends of DNA reads before analysis

Consumables - the various reagents, plates, and other materials used in the pipeline

Custom manifest - a user-created file that describes the layout and contents of a custom sample plate, either 96 or 384-well. See [File Formats](#)

DNA plate - a 384-well plate used for holding DNA from samples in the genotyping pipeline

[Back to Top](#)

E-H

Echo - an acoustic liquid handling robot used for precise dispensing of reagents in the genotyping pipeline

Echo plate - a specific (and expensive) 384-well plate used in the Echo liquid handling system for holding samples or reagents. These plates have DNA, primers, taq polymerase, water, and DNA index barcodes added to them

Exact matching - a method of sequence comparison that requires perfect identity between the query and reference sequences. It is very fast

Experiment - a collection of all the data and information related to a particular genotyping run, including samples, consumables, amplicons, and results. It exists as a subfolder with the prefix "run_", and contains all the run files, critically the "experiment.json" file, which contains the complete record of every action performed as part of the run

Fastq - a file format used for storing DNA sequencing data, which includes both the DNA sequence and its corresponding quality scores

Genotype - the genetic makeup of an organism, typically represented by a collection of specific alleles (genetic variations) present at specific loci

Genotyping - the process of determining the genetic makeup of an organism by analyzing its DNA sequence

Global alignment - a method of sequence comparison that aligns the entire length of the query sequence with the reference sequence, allowing for gaps and mismatches

[Back to Top](#)

I-L

Index barcode - a short DNA sequence added to samples for identification during sequencing.

Indexing - the process of adding index barcodes to samples for identification during sequencing

Inexact matching - a method of sequence comparison that allows for some differences between the query and reference sequences. Slower than exact matching

Information panel/viewer - a section of the interface that displays specific information about the experiment, such as loaded samples or activity logs

Local alignment - a method of sequence comparison that aligns a portion of the query sequence with the reference sequence, allowing for gaps and mismatches

Log entry - a record of an event or action taken within the experiment, including details such as time, severity level, and message

Log level - a categorization of log entries based on their severity, such as info, warning, error, etc.

[Back to Top](#)

M-P

Manifest - an input file that may contain records for more than one plate of samples

Matching - the process of comparing sequenced DNA reads to reference sequences to determine genotypes, can be exact or inexact, local or global, also known as *alignment*

Miseq - a type of sequencing platform used in the pipeline for sequencing DNA samples

Nimbus - a liquid handling robot used for plate combining and sample preparation in the pipeline. Made by the Hamilton company

NGS - Next-Generation Sequencing (e.g. with the Illumina Miseq platform)

NGSG - NGS Genotyping (Pipeline), the physical process and software application for managing and executing the genotyping process using NGS technologies

Parsing - the process of analyzing and extracting information from a file or data structure

PCR (Polymerase Chain Reaction) - a technique used to amplify specific DNA sequences, used first to amplify target regions before sequencing, and then to attach index barcodes

Plate - a flat, rectangular container with wells used for holding and organizing samples or reagents in the pipeline

Primer - a short strand of DNA that serves as a starting point for DNA synthesis during PCR amplification

Primer plate - a 384-well plate used for holding primers in the genotyping pipeline

Processes - in the context of allele matching/sequence alignment, the number of independent computational tasks being executed

[Back to Top](#)

Q-T

Quality trimming - the process of removing low-quality bases from the ends of DNA reads before analysis

Read - a sequence of DNA that has been sequenced from a sample, which is analyzed to determine genotypes

Reference sequence - a known DNA sequence used as a benchmark for comparing and aligning sequenced reads

Rodentify - a database containing information about mouse ear punch samples, including their genotypes and other metadata

Run - a specific execution of the genotyping pipeline, encompassing all the steps from sample loading to result generation, also referred to as an *experiment*

Sample - a biological specimen or aliquot that is subjected to genotyping analysis

Sequence matching - the process of comparing sequenced DNA reads to reference sequences to determine genotypes, can be exact or inexact, local or global

Short read - a brief (~100-300 base pair) DNA sequence obtained from high-throughput sequencing technologies, e.g. Illumina Miseq

Stage - a distinct phase of the genotyping pipeline, each with specific tasks and objectives (e.g. Load, Nimbus, Primers, etc.)

Tab - a subsection within a stage that organizes related tasks and information (e.g. Load Samples, Load Consumables, etc.)

Taq polymerase - a DNA polymerase enzyme used in PCR amplification to synthesize new DNA strands, loaded in 6-well plates and dispensed into Echo plates

Threads - the number of concurrent processes or tasks that can be executed simultaneously in the pipeline. Similar to *processes*

[Back to Top](#)

U-Z

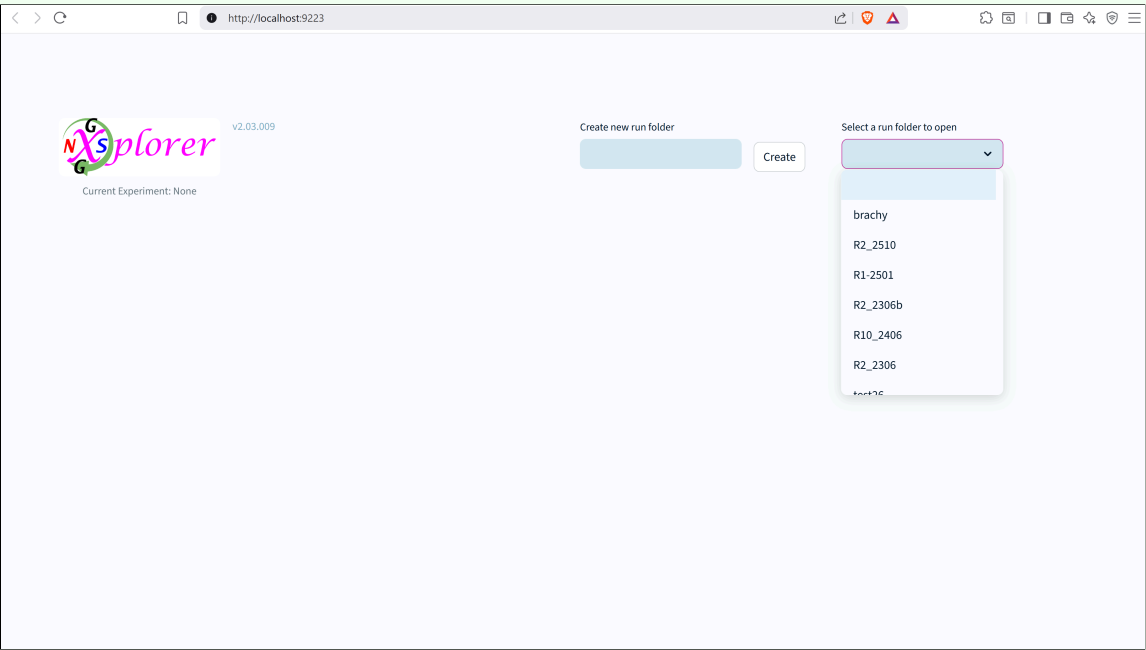
Water - provided in 6-well plates and dispensed into Echo plates. Required for PCR amplification at the Primer and Indexing stages

[Back to Top](#)

This glossary was built with the assistance of Microsoft Copilot. Please report any inaccuracies or suggestions for improvement.



NGS Explorer Start Page



Description

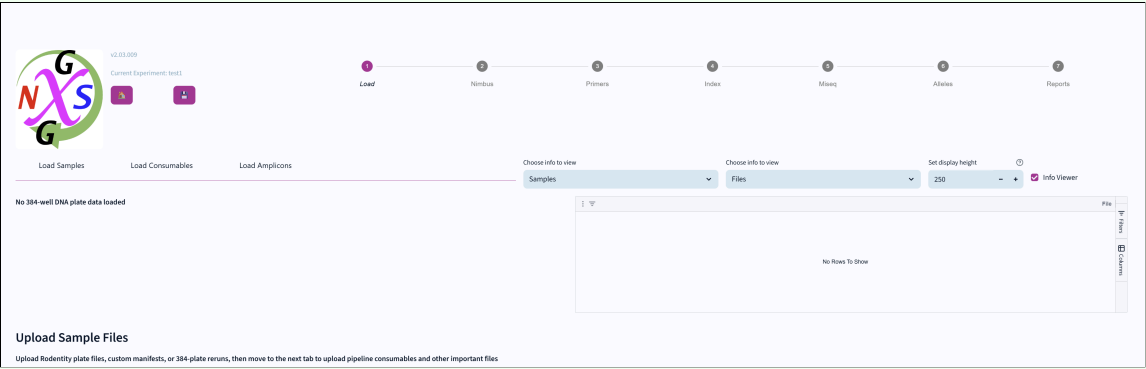
The left hand side of the start screen displays the Explorer logo, along with the version number, and the currently loaded experiment.

The right hand side of the start screen contains a field for the name of a new experiment to create, along with its button. Next to this is a drop-down menu of all existing experiments that you can load

When you make a new name for an experiment, it will be added to the list of available experiments. It will also create a new directory for the experiment files that is prepended with "run_".

NGS Genotyping Pipeline Documentation: New Experiments

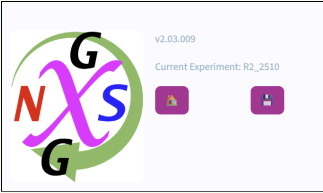
The first stage of the NGS genotyping pipeline involves loading the necessary experiment files into the system. It is the first view a user gets when they create a new experiment.



This is what a new experiment looks like when first created. Let's explain the different components.

Logo region

At the top left is the logo, version number, and currently loaded experiment name.



There are also buttons for returning to the home/start screen, and for saving the current experiment.

Saving the experiment will preserve all changes made to the experiment and allow you to reopen it at a later time. If you close the browser window without saving, you will likely lose your changes.

Pipeline widget

At the top right is a pipeline widget for navigating between different stages of the pipeline. The current stage is indicated by the right-most purple number.



Stage tabs

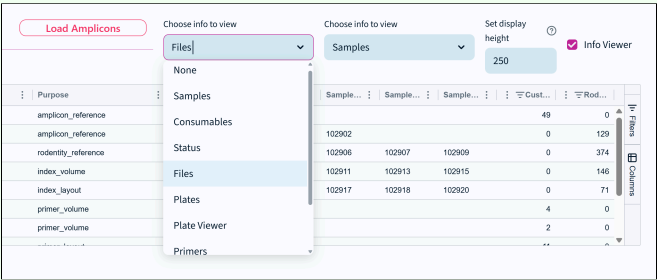
Below, and to the left are page tabs for different sections of the load stage.



The **Load** stage has three tabs: Load Samples, Load Consumables, Load Amplicons

Information panels

To the right, and below the tabs, are the controls and display areas for a pair of information panels.



You can [read more about the various information panels](#), but for now you should know that these viewers default to panels considered most appropriate for the selected stage and tab.

Appropriate panels can be used to select files and plates that have been loaded, and remove them from the experiment.

It is also possible to adjust the height of the viewer, or turn it off completely with the checkbox on the far right of the screen.

Stage- and tab-specific content

Each stage and tab has its own specific content and controls. The Load stage has three tabs, and each tab will display different file upload options

Upload Sample Files

Upload Rodentity plate files, custom manifests, or 384-plate reruns, then move to the next tab to upload pipeline consumables and other important files

Upload Rodentity Ear Punch Plates

Add up to four Rodentity JSON ear-punch plate (96-well) files at a time and click on **Submit**. This will move the plate identifiers into the four slots below (P1-4).

Choose up to 4 Rodentity JSON files to fill available DNA plate positions

Drag and drop files here

Limit 200MB per file • JSON

Browse files

Submit

Then assign a DNA plate (384-well) barcode identifier for plate combining with the Hamilton Nimbus, and click on **Accept**

P1:

P2:

P3:

P4:

Clear IDs

Destination plate ID (barcode)

Accept

For example, here is the file upload interface to Rodentity Ear Punch Plates, specific to the Load Samples tab.

You can find out how each of these works in the appropriate stage documentation.

Save your progress!

Remember to save

Don't forget to save your progress regularly! Saving will preserve all changes made to the experiment and allow you to reopen it at a later time.

If you close the browser window without saving, you will likely lose your changes. That includes any files that have been uploaded or generated using the pipeline.

The pipeline is designed to be rigorous and reliable, so if it detects files in the experiment folder with an expected name, that aren't recorded in the experiment records, it will delete them.

Activity log

At the bottom of every page is another information panel displaying the activity log.

Choose info to view

Log

Choose info to view

None

Set display height

350

Time	Level	Message	Function name	Func line
2025-11-20 12:41	Success	created primer assay families file run_R2_2510primers.csv	generate_primer_assayfams	1181
2025-11-20 12:41	Success	Created reference sequences file run_R2_2510targets.fa containing 882 sequences	generate_targets	1149
2025-11-04 15:11	Info	C:\Users\Cami\NGS\genovenv\ngs\Scripts\python.exe bin\bingmatch.py --ncpus 8 --outdir run_R2_2510 --margin 0.9 --ide...	main	1104
2025-11-04 15:11	Success	created primer assay families file run_R2_2510primers.csv	generate_primer_assayfams	1181
2025-11-04 15:11	Success	Created reference sequences file run_R2_2510targets.fa containing 882 sequences	generate_targets	1149
2025-10-30 13:32	Info	C:\Users\Cami\NGS\genovenv\ngs\Scripts\python.exe bin\bingmatch.py --ncpus 8 --outdir run_R2_2510 --margin 0.9 --ide...	main	1106
2025-10-30 13:32	Success	created primer assay families file run_R2_2510primers.csv	generate_primer_assayfams	1181
2025-10-30 13:32	Success	Created reference sequences file run_R2_2510targets.fa containing 882 sequences	generate_targets	1149
2025-10-30 13:27	Info	C:\Users\Cami\NGS\genovenv\ngs\Scripts\python.exe bin\bingmatch.py --ncpus 8 --outdir run_R2_2510 --margin 0.9 --ide...	main	1106
2025-10-30 13:27	Success	created primer assay families file run_R2_2510primers.csv	generate_primer_assayfams	1181
2025-10-30 13:27	Success	Created reference sequences file run_R2_2510targets.fa containing 882 sequences	generate_targets	1149

This is a crucial tool for tracking all actions taken within the experiment. It is possible to change the view from the log panel to another information panel type, and you can have a second panel visible too. By default only the log is shown here to provide sufficient horizontal screen space.

The columns in the log display are (from left to right): Time, Level, Message, Function name, Func line.

- Time - the time and date the log entry was created
- Level - the severity level of the log entry (e.g. info, warning, error)
- Message - a descriptive message about the log entry
- Function name - the name of the function where the log entry was generated
- Func line - the programming line number in the function where the log entry was generated

Message levels are used to indicate the severity of an issue.

- Info - informational messages providing general updates on events
- Success - indicates successful completion of an action or stage
- Failure - indicates a failed action or stage. These are not typically problematic
- Warning - indicates a potential issue that may not stop the pipeline from running, such as insufficient primer volumes, but should be considered
- Error - indicates an issue that will prevent the pipeline from running, such as missing files or incorrect file formats
- Critical - indicates a severe issue that requires immediate attention, such as system failures or data corruption. These indicate that the pipeline is in an inconsistent state or has encountered a programming error. You should raise any critical issues immediately.
- Debug - detailed information for debugging purposes - unlikely to be seen in regular use

To aid users in spotting important issues, **warnings** are displayed in yellow, **errors** are displayed in red, and **critical** messages are in purple.

You cannot remove or clear log entries.

Under the upload button is a set of four drop-down selectors to let the user specify up to four plates to combine into a single 384-well "DNA" plate. To the right is a text box for typing in a unique plate ID for the combined "DNA" plate, to be created using the Hamilton Nimbus robot.

Custom 384-Well Plates

Upload 384-well Plate Manifest Files

You may add user-defined 384-well sample (DNA) plates here, for reruns or custom analysis.

Upload 384-well custom manifests - CSV or XLIX

Drag and drop files here
Limit 200MB per file • CSV, XLIX

Browse files

Upload Files

Remember to save

The main input widget for working with custom 384-well plates is shown above. Since these are already in the 384-well format, they do not need to be combined using the Hamilton Nimbus robot.

At the bottom right you will see a text button encouraging you to save your progress. If you close the browser window without saving, you will lose your progress.

Load Consumables

10.01.2023

Current Experiment: RG_2010

Load Samples

Load Consumables

Load Amplicons

Choose info to view

Consumables

Samples

Choose info to view

Set display height

250

Info Viewer

Program	BarcodeID	Wells	Type	Entries	DNA1...	Sample...	Sample...	Sample...	Wells...
lapview (PCR 1)	1254	6	69C5_A2_BF2	6	20100107	102000	102002	102007	0
lapview (PCR 2)	2545	6	69C5_A2_BF2	6	20100107	102000	102002	102007	0
primer	PT5	384	384FP_A2_BF	384	2010000	102005	102006	102009	0
primer	PT5	384	384FP_A2_BF	384	2010000	102010	102011	102013	0
index	3400	384	384FP_A2_BF	384	2010010	102010	102017	102018	0
fluorescence map	run_R2_2010y4b...	0	File	0	2010001				0
amplification map	run_R2_2010y4b...	0	File	0	2010002				0

Here we can see an example of an existing experiment with consumables loaded. The default viewer panels are Consumables and Files.

There are four widgets for uploading/setting consumable information (from top to bottom): Reference/Assay files, Primer files, Index files, Custom volume transfers

Reference/Assay Files - see the File Formats documentation page for details on the required formats.

Upload Reference/Assay Files

Drag and drop files here
Limit 200MB per file • CSV

Browse files

Upload Reference/Assay Files

Drag and drop files here
Limit 200MB per file • CSV

Browse files

Remember to save

Primer Files - layout and volume files are needed for each primer plate. See the File Formats documentation page for details.

Upload Primer Files

Drag and drop files here
Limit 200MB per file • CSV

Browse files

Upload Primer Files

Drag and drop files here
Limit 200MB per file • CSV

Browse files

Remember to save

Index Files - layout and volume files are needed. These should be standard file used in every run, but see the File Formats documentation page for details.

Upload Index Files

Drag and drop files here
Limit 200MB per file • CSV

Browse files

Upload Index Files

Drag and drop files here
Limit 200MB per file • CSV

Browse files

Remember to save

Custom Volume Transfers - do not adjust these unless specifically instructed. Changes are noted in the log.

Custom Volumes

Reference for DNA and Primers

Make a table of how many times each primer volume must be added (Current state: 200)

Upload Reference Files

Drag and drop files here
Limit 200MB per file • CSV

Browse files

Upload Reference Files

Drag and drop files here
Limit 200MB per file • CSV

Browse files

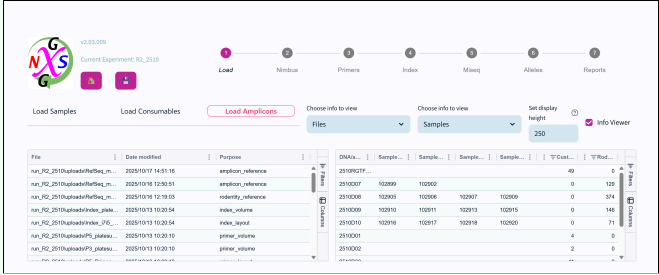
Remember to save

Primer	Volume	Transfer	Volume	Transfer	Volume	Transfer	Volume	Transfer
100	200	100	200	100	200	100	200	100

And don't forget to save your pipeline progress!

Remember to save

Load Amplicons



Here we can see an example of an existing experiment with amplicons loaded. The default viewer panels are Samples (including Amplicons) and Files.

Upload Amplicon Files

Upload amplicon plates and reference files

Upload Amplicon Plate Files

You may add extra 384-well plates of pre-prepared amplicons. These will be incorporated during indexing (stage 4)

Upload Data Amplicon Plates - CSV or XLSX (validates MiSeq and Stage3 CSV files if they exist)

Drag and drop files here
Limit 200MB per file - CSV, XLSX

Browse Files

Upload Files

Upload Amplicon Sequence Reference Files

Drag and drop files here
Limit 200MB per file - TXT, FASTA

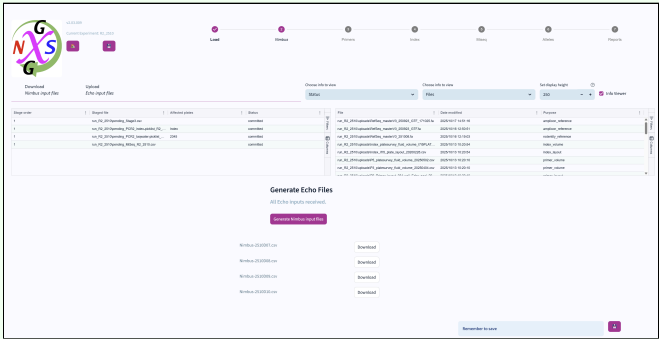
Browse Files

Upload Files

Any number of 384-well amplicon manifest files can be uploaded at once. Each amplicon reference file uploaded can be selected separately later to customise amplicon matching (in the Allele stage).

Stage 2: Hamilton Nimbus Robot

The second stage of the NGS genotyping pipeline involves using the Hamilton Nimbus robot to perform plate aggregation from 96-well to 384-well format.

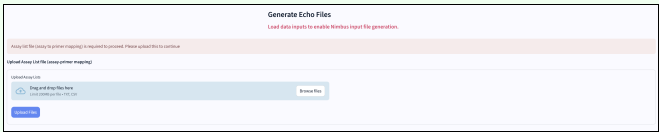


After loading the necessary files in the Load stage, you can download the Hamilton Nimbus robot input files from the Nimbus stage page.

These files contain all the information needed for the robot to perform the plate aggregation step.

You will need to generate the appropriate Nimbus inputs files by pressing the purple button labeled "Generate Nimbus input files". If files already exist then you will be asked whether you wish to overwrite them or not.

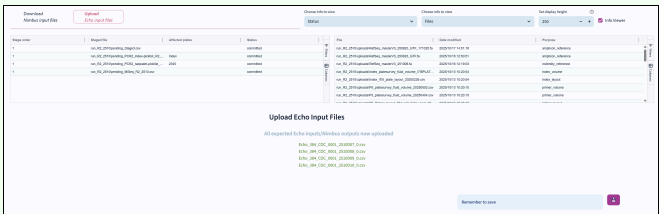
If you haven't loaded the necessary files in the Load stage, you will see a message indicating this:



You are prompted to load the necessary files before proceeding, and a widget for uploading the assay-primer map file is made available if not yet loaded.

While the assay-map is not strictly required for the Nimbus robot, this information is needed to correctly generate the 384-well "DNA" plate records that are used as inputs for the next stage of the pipeline.

After you have stepped away to run the Hamilton Nimbus robot, you can upload the resulting Echo_384_COC_ files back into the system.



These files contain the information about how the samples have been aggregated into 384-well format, and are necessary for the subsequent stages of the pipeline. The pipeline combines them with the information it already has to generate an internal representation of the 384-well "DNA" sample plates that now exist physically.

For each Nimbus upload file created, the pipeline will expect to see a corresponding Echo_384_COC_file uploaded, and will prompt you for each of them, though you can proceed to the next stage of the pipeline as long as one Echo_384_COC file has been uploaded.

The third stage of the NGS genotyping pipeline involves performing primer PCR to amplify the target regions.

Samples, taq polymerase, and water, and PCR primers, are added to the wells of the Echo plates.

[illegible]

Users can see the quantities and details of each component, and enable/disable them using checkboxes.

Add Barcodes

1 Volume Barcode

Type in the barcode for this volume

Save

1 Local Book Plate Barcode

Type in the code for this bookplate

Save

Upload Files

Upload a new file (PDF format)

Upload the file path of the file name is the same for topics and volumes (e.g. FL_Spanish_23_volume.pdf)

Upload from your device

Drop files here to upload

Choose files

Upload from your device

Drop files here to upload

Choose files

Custom Volumes

Accession to 100 and beyond

Doc. Access: Primary Tagset Primary Volume values must total 20000. Current total: 2000

Current volume(s): Edit current volumes

Search to default

DOC_ID	PRIMARY_VOL	SP	PRIMARY_VOL_2	PRIMARY_VOL_3	SECOND_VOL	THIRD_VOL	FOURTH_VOL	FIFTH_VOL	SIXTH_VOL	SEVENTH_VOL	EIGHTH_VOL	NINTH_VOL	TENTH_VOL
000001	001												

Accession to 100 and beyond

Below that, there are upload widgets for primer layout and volume files.

Lastly, there is a widget for adjusting liquid transfer volumes with the Echo. Do not adjust these values unless instructed!

And don't forget to save your progress!

Generate Primer (PCR 1) Echo Picklists

Clicking on a sequence name will lead to a page for that primer. PCR 1 and 2 are click on the Reference RNA Picklists below. Picklists below are for use in the primer design tool.

RNA Picklists

- 20180001
- 20180002
- 20180003
- 20180004
- 20180005
- 20180006
- 20180007
- 20180008
- 20180009
- 20180010
- 20180011
- 20180012
- 20180013
- 20180014
- 20180015
- 20180016
- 20180017
- 20180018
- 20180019
- 20180020
- 20180021
- 20180022
- 20180023
- 20180024
- 20180025
- 20180026
- 20180027
- 20180028
- 20180029
- 20180030
- 20180031
- 20180032
- 20180033
- 20180034
- 20180035
- 20180036
- 20180037
- 20180038
- 20180039
- 20180040
- 20180041
- 20180042
- 20180043
- 20180044
- 20180045
- 20180046
- 20180047
- 20180048
- 20180049
- 20180050
- 20180051
- 20180052
- 20180053
- 20180054
- 20180055
- 20180056
- 20180057
- 20180058
- 20180059
- 20180060
- 20180061
- 20180062
- 20180063
- 20180064
- 20180065
- 20180066
- 20180067
- 20180068
- 20180069
- 20180070
- 20180071
- 20180072
- 20180073
- 20180074
- 20180075
- 20180076
- 20180077
- 20180078
- 20180079
- 20180080
- 20180081
- 20180082
- 20180083
- 20180084
- 20180085
- 20180086
- 20180087
- 20180088
- 20180089
- 20180090
- 20180091
- 20180092
- 20180093
- 20180094
- 20180095
- 20180096
- 20180097
- 20180098
- 20180099
- 20180100
- 20180101
- 20180102
- 20180103
- 20180104
- 20180105
- 20180106
- 20180107
- 20180108
- 20180109
- 20180110
- 20180111
- 20180112
- 20180113
- 20180114
- 20180115
- 20180116
- 20180117
- 20180118
- 20180119
- 20180120
- 20180121
- 20180122
- 20180123
- 20180124
- 20180125
- 20180126
- 20180127
- 20180128
- 20180129
- 20180130
- 20180131
- 20180132
- 20180133
- 20180134
- 20180135
- 20180136
- 20180137
- 20180138
- 20180139
- 20180140
- 20180141
- 20180142
- 20180143
- 20180144
- 20180145
- 20180146
- 20180147
- 20180148
- 20180149
- 20180150
- 20180151
- 20180152
- 20180153
- 20180154
- 20180155
- 20180156
- 20180157
- 20180158
- 20180159
- 20180160
- 20180161
- 20180162
- 20180163
- 20180164
- 20180165
- 20180166
- 20180167
- 20180168
- 20180169
- 20180170
- 20180171
- 20180172
- 20180173
- 20180174
- 20180175
- 20180176
- 20180177
- 20180178
- 20180179
- 20180180
- 20180181
- 20180182
- 20180183
- 20180184
- 20180185
- 20180186
- 20180187
- 20180188
- 20180189
- 20180190
- 20180191
- 20180192
- 20180193
- 20180194
- 20180195
- 20180196
- 20180197
- 20180198
- 20180199
- 20180200
- 20180201
- 20180202
- 20180203
- 20180204
- 20180205
- 20180206
- 20180207
- 20180208
- 20180209
- 20180210
- 20180211
- 20180212
- 20180213
- 20180214
- 20180215
- 20180216
- 20180217
- 20180218
- 20180219
- 20180220
- 20180221
- 20180222
- 20180223
- 20180224
- 20180225
- 20180226
- 20180227
- 20180228
- 20180229
- 20180230
- 20180231
- 20180232
- 20180233
- 20180234
- 20180235
- 20180236
- 20180237
- 20180238
- 20180239
- 20180240
- 20180241
- 20180242
- 20180243
- 20180244
- 20180245
- 20180246
- 20180247
- 20180248
- 20180249
- 20180250
- 20180251
- 20180252
- 20180253
- 20180254
- 20180255
- 20180256
- 20180257
- 20180258
- 20180259
- 20180260
- 20180261
- 20180262
- 20180263
- 20180264
- 20180265
- 20180266
- 20180267
- 20180268
- 20180269
- 20180270
- 20180271
- 20180272
- 20180273
- 20180274
- 20180275
- 20180276
- 20180277
- 20180278
- 20180279
- 20180280
- 20180281
- 20180282
- 20180283
- 20180284</

After confirming that the selected components are correct and enabled with tickboxes, users can generate the picklists for the PCR reactions.

Any there are missing primers, they will be indicated in the summary table.

Ticking the "Force picklist generation" checkbox will override any warnings and generate the picklists regardless.

If any of the files to be generated already exist, the user will be prompted to confirm overwriting them.

The fourth stage of the NGS genotyping pipeline involves performing a second round of PCR to add unique index barcodes to the DNA fragments. As with the Primer stage, there are two tabs for PCR Components and Generating Picklists.

PCR2 Components

You can see all of the component plates available for indexing [here](#).

You can add plates specific to the index PCR stage here, namely dedicated taq/water plates for PCR2, index layout and volume, or extra amplicon plates.

There is no need to add additional PCR plates here, it's simply that the widget offers both the PCR and Taq/water barcode types.

Lastly, you can set custom transfer volumes for the index (PCR2) reaction. Do not do this unless specifically instructed!

Generate Picklists

Just like the Primer stage, this tab allows you to generate picklists for the index PCR reactions.

There are checkboxes to select the various available components for this pipeline stage.

There is no "force" option for generating files at this stage as the index barcodes are all required to be present at full volumes.

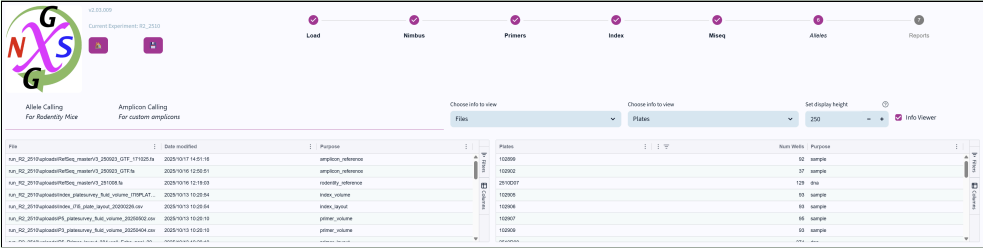
If the Echo PCR2 picklists already exist, you can choose to overwrite them or keep the existing ones.

...and don't forget to save your progress!



Stage 6: Allele Calling and Counting

The sixth stage of the NGS genotyping pipeline involves calling and counting alleles from the sequencing data.



There are two tabs at the Alleles stage - "Allele Calling for Rodentia Mice" and "Amplicon Calling for custom amplicons"

The upper info viewer panels default to Files and Plates as the most relevant items to track, but it's unlikely to be very useful in practice here.

Allele Calling for Rodentia Mice

This is the main NGS Genotyping pipeline tab for calling (identifying and counting) sequences for each sample/assay well.

Available FASTQ files in RAW folder

1. run_R2_2510\raw\2510R2PCR1_A01_S1_L001_R1_001.fastq.gz, run_R2_2510\raw\2510R2PCR1_A01_S1_L001_R2_001.fastq.gz

...

1,410. run_R2_2510\raw\2510RGTFPCR1_P03_S1374_L001_R1_001.fastq.gz, run_R2_2510\raw\2510RGTFPCR1_P03_S1374_L001_R2_001.fastq.gz

Count sequences

The first widget displays the first and last (alphanumerically ordered) FASTQ file that have been manually copied to the "raw" subfolder within the experiment/run folder.

There is also a button here which will launch a count all the available raw sequences in the FASTQ files, which is a useful indicator of overall sequencing success.

It may take a few minutes to count all the sequences seen in a large genotyping run with thousands of files. Here is an example of the counts progressing:

Created reference sequences file run_R2_2510\targets.fa containing 882 sequences

created primer assay families file run_R2_2510\primers.csv

Available FASTQ files in RAW folder

1. run_R2_2510\raw\2510R2PCR1_A01_S1_L001_R1_001.fastq.gz, run_R2_2510\raw\2510R2PCR1_A01_S1_L001_R2_001.fastq.gz

...

1,410. run_R2_2510\raw\2510RGTFPCR1_P03_S1374_L001_R1_001.fastq.gz, run_R2_2510\raw\2510RGTFPCR1_P03_S1374_L001_R2_001.fastq.gz

Count sequences

Reading 122/1410

You can see in this example that there are also messages announcing that the "targets.fa" and "primers.csv" files have been created. These are required by the allele calling process.

Select references to match against

run_R2_2510\uploaddir\RefSeq_masterV3_251008.fa

Number of processes to run simultaneously, default: 8

8

Require lengths of read sequences and target references to be proportionally similar by this amount. Value must be between 0.0 and 1.0, default: 0.9

0.9

Proportion of identity required for nearest match. default: 0.9. Must be between 0.0 and 1.0

0.9

Do not match unique sequences with less than this many reads coverage, default: 5

5

Do not match unique sequences with less than this proportion of the reads seen for the most observed (expected) allele, default: 0.1. Must be between 0.0 and 1.0

0.1

☐ Enable intersect matching

☐ Exhaustive mode: try to match every sequence, no matter how few counts

☐ Turn on debugging for allele calling

Run allele calling

Next is the panel of options for the allele calling process. Before we discuss each option, here's a quick explanation about what happens when the "Run allele calling" button is pressed

1. All paired-end FASTQ sequence record (two files per well) are trimmed of low quality bases, and short reads are filtered out
2. Clean paired sequences are merged. Unmerged reads are saved, but not included in later processing
3. Merged sequences are matched against the reference sequences expected for them given the primers that were used

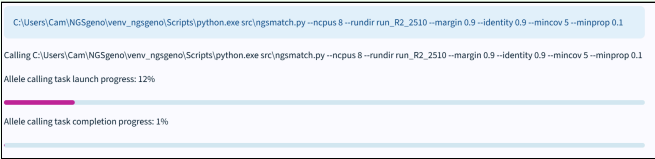
- With that in mind, let's go over these options:

- The values for these parameters reset to the defaults after every calling run.

If you are still unclear about the meaning of any of these settings you can read the [technical details](#) page for allele calling/matching, but it's probably better to trust the defaults

When you are ready, click the button labelled "Allele calling"

You will soon see a progress bar widget that displays both the assignment of tasks to available computational processes, and another bar showing the actual task completion.



Depending on your settings it can take a few minutes to exact match a full set of mouse genotypes, and 10-15 minutes if also inexact matching.

Amplicon Calling for custom amplicons

Amplicon calling is slightly different from regular genotyping in that it always uses inexact matching to let clients explore the variety of sequences that may be present across an amplicon locus.

The interface is also designed around an expectation that there could be amplicon plates present from different clients simultaneously, each requiring a different reference to match against.

Amplicon Calling

This section is for custom amplicon calling. Amplicon sequences are always inexact matched against their expected primer to find variants. Please choose the amplicon reference you wish to use or upload a new one

Upload Amplicon Sequence Reference Files

 Drag and drop files here
Limit 200MB per file • TXT, FA, FASTA

Browse files

Upload Files

The first widget here is for uploading an extra amplicon reference files that might be needed for matching.

Select amplicon plates for allele calling

Amplicon Plates

☐ 2510RGTFPCR1

No amplicon plates selected

Select amplicon references to match against

Amplicon reference Files

☐ RefSeq_masterV3_250923_GTF.fa

☐ RefSeq_masterV3_250923_GTF_171025.fa

No amplicon references selected

Next users can chose the combination of amplicon plates and references that are to be matched.

Amplicon matching options are essentially the same as for the genotyping workflow, with the only real change being that inexact matching is always performed for alleles.

Please refer to the Allele calling tab detailed above for descriptions of the calling options.

Number of processes to run simultaneously, default: 8

8 - +

Require lengths of read sequences and target references to be proportionally similar by this amount. Value must be between 0.0 and 1.0 default 0.9

0.9 - +

Proportion of identity required for inexact match, default 0.9. Must be between 0.0 and 1.0

0.9 - +

Do not match unique sequences with less than this many reads coverage, default 5

5 - +

Do not match unique sequences with less than this proportion of the reads seen for the most observed (expected) allele, default 0.1. Must be between 0.0 and 1.0

0.1 - +

☐ Exhaustive mode: try to match every sequence, no matter how few counts

☐ Turn on debugging for allele calling

Run amplicon calling

Once you have set an parameters required, click on "Run amplicon calling" to launch the processing of amplicon sequences.

The progress bars are the same as for the genotyping workflow:

Created reference sequences file run_R2_2510\amplicon_targets.fa containing 3 sequences

C:\Users\Cam\NGSgeno\env\ngsgeo\Scripts\python.exe src\ngsmatch.py --ncpus 8 --rundir run_R2_2510 --mincov 5 --minprop 0.1 --amplicons p2510RGTFPCR1p --targets amplicon_targets.fa

Calling C:\Users\Cam\NGSgeno\env\ngsgeo\Scripts\python.exe src\ngsmatch.py --ncpus 8 --rundir run_R2_2510 --mincov 5 --minprop 0.1 --amplicons p2510RGTFPCR1p --targets amplicon_targets.fa

Allele calling task launch progress: 100%

Allele calling task completion progress: 0%

Those with sharp eyes will notice some differences in the command line that is displayed compared to the genotyping command line

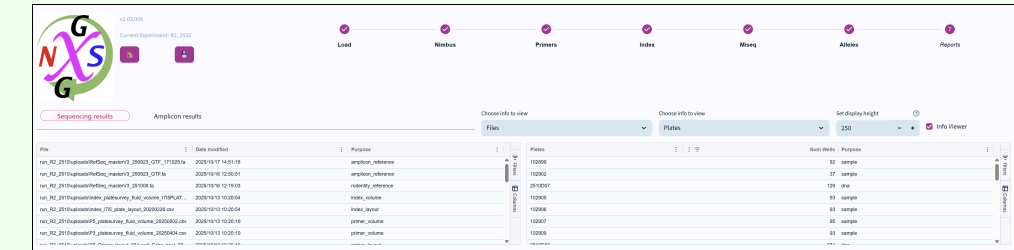
Each amplicon plate may take approximately one minute to process



Stage 7: Reporting

The seventh stage of the NGS genotyping pipeline displays reduced column summary tables of results from the Allele stage.

For the Amplicon pipeline, it also allows multiple sequence alignments to be viewed, filtered for relevance, and then saved to a PDF document.



There are two tabs: 1) "Sequencing Results" for Rodentity, Custom and "Other" (amplicons that might be from Rodentity, or otherwise use the standard pipeline), and 2) "Amplicon Results" for Amplicons plates that were run through the Amplicon workflow.

The info viewer is set to show Files and Samples, but likely users will want to turn it off to focus on the result tables.

Sequencing Results

This tab displays the sequencing results for Rodentity, Custom, and "Other" samples.

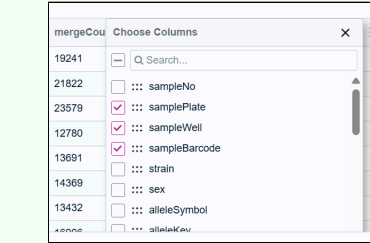
Unreported rodentity											
samplePlate	sampleWell	sampleBarcode	primer	readCount	mergeCount	seqCount	seqName	efficiency	otherCount	otherName	filterProportion
12888	A3	MA327200032	U6a-wb-F862	19111	19241	8592.4108	U6a-wb-F862_MUT-wb-D3...	0.594			0.18
12888	A4	MA327200032	U6a-wb	22996	21922	12842	U6a-wb_MUT-wb-F1218F	0.589	8605.76.76.76.76.76.76...	GCTCCGATTTTTCAGTCACTACT...	0.18
12888	A5	MA327200032	Tg9aG4181_BT	24528	23979	18127	Tg9aG4181_BT	0.589			0.18
12888	A6	MA327200032	Tg9aG4181_BT	13437	12780	8073	Tg9aG4181_BT	0.671			0.18
12888	A5	MA327200032	U6a-wb	14129	13641	3663.3004	U6a-wb_MUT-wb_MUT-wb...	0.627			0.18
12888	A6	MA327200032	Pg9a	12148	12681	7362	Pg9a_MUT-wb	0.647			0.18
12888	A6	MA327200032	Tg9aG4181_BT	14336	14432	8428	Tg9aG4181_BT	0.627			0.18
12888	A7	MA327200032	Tg9aG4181_BT	21444	21444	12444	Tg9aG4181_BT	0.644			0.18
Unreported custom											
samplePlate	sampleWell	sampleBarcode	primer	readCount	mergeCount	seqCount	seqName	efficiency	otherCount	otherName	filterProportion
2010001	B8	MA326800036	U6a-wb	27012	26593	13897	U6a-wb_MUT-wb-F1218F	0.589	105.18.85.85.85.85.85...	GCTCCGATTTTTCAGTCACTACT...	0.18
2010001	B12	MA326800036	U6a-wb	20048	19251	8472.2847	U6a-wb_MUT-wb-F1218F...	0.589			0.18
2010001	B3	MA326800036	U6a-wb	2114	2024	9126.9	U6a-wb_MUT-wb-F1218F...	0.589	207.147.83.83.83.83.83...	Pag1-wb_MUT-wb-MUT-wb...	0.18
2010001	B3	MA326800036	U6a-wb	1688	1688	25.14.14	U6a-wb_MUT-wb-F1218F...	0.61	162.18.18.18.18.18.18...	Tag1-wb_MUT-wb-MUT-wb...	0.18
2010001	B3	MA326800036	U6a-wb	1391	1324	46.27.27	U6a-wb_MUT-wb-F1218F...	0.589	204.75.48.48.48.48.48...	GCTCCGATTTTTCAGTCACTACT...	0.18
2010001	B3	MA326800036	U6a-wb	14659	14439	8913	U6a-wb_MUT-wb	0.61			0.18
2010001	L12	MA326800036	U6a-wb	22775	21925	9124.4703	U6a-wb_MUT-wb-F1218F...	0.603			0.18

Each table says "Unreported" in the header, but that's because this pipeline simply have that need. Let's look at the columns in the table.

- samplePlate**: the name of the sample, as entered in the Load stage
- sampleWell**: the well that the sample was loaded into in the Load stage
- sampleBarcode**: the barcode of the sample, as entered in the Load stage
- primer**: the primer used for the sequencing reaction
- readCount**: the number of raw reads observed for the sample
- mergeCount**: the number of merged reads observed for the sample
- seqCount**: semicolon separated list of the number of merged reads matched against an expected and recognised reference sequence
- seqName**: semicolon separated list of names of the reference sequences, corresponding to the counts in the previous column
- efficiency**: the efficiency of the sequencing reaction, as measured by the proportion of reads assigned to the expected reference sequences for the given primer
- otherCount**: semicolon separated list of the number of reads that did not match an expected reference sequence
- otherName**: semicolon separated list of names or merged sequences that did not match the expected primer options, but were still seen in significant quantities. Note that "Other" refers to the collection of all other sequences that were seen in insignificant quantities individually, but the sum total is large enough to be significant. The number of different observed sequences is given in parentheses
- filterProportion**: a proportional cutoff for the listing of observed sequences

Tables can be ordered by any column (just click on them), and can be ascending or descending.

It is important to remember that this table shows many less columns of information than are actually available to the user.



By right-clicking on the column headers you can customise the table to display different column choices. Here is a list of the columns that aren't displayed by default:

- sampleNo**: the sample number, as assigned at indexing
- strain**: the organismal strain of the sample
- sex**: the sex of the sample
- alleleSymbol**: from Rodentity (text)
- alleleKey**: from Rodentity (numeric)
- assayKey**: from Rodentity (numeric)
- assays**: from Rodentity (text)

8. **assayFamilies**: linking assays and primers for a complex genotype
9. **clientName**: the name of the client associated with the sample
10. **sampleName**: a descriptive name for the sample
11. **dnaPlate**: the name of the DNA plate to which the sample belongs
12. **dnaWell**: the well in the DNA plate to which the sample belongs
13. **primerPlate**: the plate containing the primer associated with this sample entry
14. **primerWell**: the corresponding well in the primerPlate for the given primer
15. **pcrPlate**: the final Echo PCR plate into which the specific combination of sample and primer were dispensed
16. **pcrWell**: the well in the PCR plate used by the this sample/primer combination
17. **i7bc**: the i7 forward index barcode
18. **i7name**: the name associated with the i7 forward barcode
19. **i7well**: the well for the i7 forward index used with this sample
20. **i5bc**: the i5 reverse index barcode
21. **i5name**: the name associated with the i5 reverse barcode
22. **i5well**: the well for the i5 reverse index used with this sample
23. **index_plate**: the plate barcode containing the i7 and i5 index barcodes
24. **cleanCount**: the number of reads passing base quality and length trimming

The screenshot shows a context menu titled "Unreported custom:" with the following options: "samplePlate", "sampleWell", and "sa". Below these, a table is visible with columns "2510D02" and "L9". The table contains several rows of data. A right-click context menu is open over the table, showing options: "Cut" (Ctrl+X), "Copy" (Ctrl+C), "Copy with Headers", "Copy with Group Headers", "Paste" (Ctrl+V), "Export" (highlighted with a red box), "CSV Export", and "Excel Export".

Amplicon Results

It has its own tab because it supports additional features specific to the Amplicon workflow. Notably the ability to perform multiple sequence alignments of the observed sequences. These can filtered out using the adjustable proportion filter (this is the only column that supports changing the values in each row).

[illegible]

Clicking on a row to select it will open an additional table underneath that show the variants (if any) that pass the `filtProportion` cutoff.

These are sent to multiple sequence alignment and displayed (in non-monospace font due to graphics restrictions). These can be further down-selected using the checkboxes at left, and a description added in the available text box.

Then clicking "Make report entry" will add them to a PDF report file (named `amplicon_alignments.pdf`) containing aligned sequences for each chosen sample.

Once a sample has been added to the report, it is removed from the "Unreported amplicon" table and placed in a new table at bottom named "Reported amplicons".

Currently (v2.04.000) it isn't possible to remove individual amplicons from the report PDF, but you can choose to reset the amplicon report PDF using the "Reset amplicon report" button underneath this table.



Amplicon Report

Sequence variant

0 GTF-Stag2-KO_WT
1 49C/T61A/C63G/T

Sequence

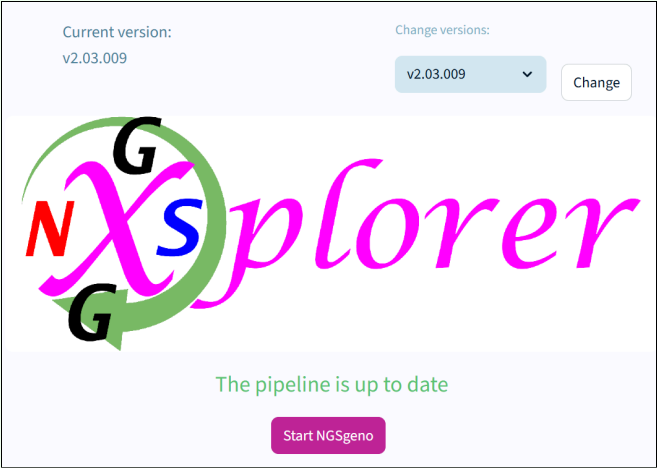
0 AACACACAAAGACAGTACGAGGCAGAACGGAATAAAATGATTGGAAAACGAGCTAACGAAAGGCTAGAACTCCTGTTACAAAAGCGTAAAGAGGTAAGCT
1 AACACACAAAGACAGTACGAGGCAGAACGGAATAAAATGATTGGAAAATGAGCTAACGAAAGGCTAGAACTCCTGTTACAAAAGCGTAAAGAGGTAAGCT
^1 ^20 ^40 ^60 ^80 ^100

0
1 ^101 ^120 ^140 ^160 ^180 ^200

This is a sample of the report PDF that is generated. Each sample gets its own page for clarity. Standard usage is to copy these into another document for final markup.



NGSG Version Select allows the user to check for new pipeline updates and choose the version they wish to run.



Make sure that you have the Ethernet cable plugged in before starting!

When you run this, it will look for new versions and pull them down to the workstation computer.

You can then choose which version you wish to use and launch it directly from within the application.



NGS Genotyping Pipeline Documentation: NGSG Reference

The NGSG Reference web tool allows users to convert two-column MS Excel spreadsheets of reference names and sequences to FASTA format. It uses a pipeline of format checkers to report any deviations (unexpected symbols for instance) before a final conversion is made.

NGS Genotyping: Reference converter

Upload an Excel document containing references in two columns (name and sequence)

Choose an Excel reference sequence file (XLSX format)

 Drag and drop file here
Limit 200MB per file

Browse files

Once all formating issues are resolved, a FASTA format file of the same name but with the .fa extension will be saved to the Downloads folder.



The NGSG Retype application is a standalone software tool that helps users to select failed assays/samples/primer-pairs for rerunning.

NGS Genotyping: Retype failed assays

Upload a genotyping report file (results workbook) in the left hand panel, and the matching Stage3.csv for this run in the right hand panel

Choose a genotyping report file

Drag and drop file here

Limit 200MB per file

Browse files

No failed assays seen

Clear session

You are first required to use the widget on the left of the screen to upload the "results_workbook.csv" file that comes back from the Rodentity server. See the file formats page for more details about this.

After that has been successfully read, a new widget will appear on the right hand side of the screen asking for you to upload the Stage3.csv from the matching experiment (run) folder.

Once this has been done, Retype will create a custom 384-well manifest file that will inform the Echo liquid handler which DNA plates and wells it needs to rerun these failed samples-primer combinations.

Choose a genotyping report file

Drag and drop file here

Limit 200MB per file

Browse files

results_workbook.csv

12.4KB

X

The following sample/assay combinations appear to have failed from the run:

Barcode: M035658200149 plate: 96990 well: B8 assays: R26-tdTomato allele_keys: Gt(ROSA)26Sor:TdTom

Barcode: M035658300150 plate: 96990 well: C8 assays: R26-tdTomato allele_keys: Gt(ROSA)26Sor:TdTom

Barcode: M035659700152 plate: 96990 well: E8 assays: Pvalb-cre; R26-tdTomato allele_keys: Pvalb:Cre; Gt(ROSA)26Sor:TdTom

Barcode: M035659800153 plate: 96990 well: F8 assays: Pvalb-cre; R26-tdTomato allele_keys: Pvalb:Cre; Gt(ROSA)26Sor:TdTom

Barcode: M035659900154 plate: 96990 well: G8 assays: R26-tdTomato; Pvalb-cre allele_keys: Gt(ROSA)26Sor:TdTom; Pvalb:Cre

Barcode: M035660000155 plate: 96990 well: H8 assays: R26-tdTomato; Pvalb-cre allele_keys: Gt(ROSA)26Sor:TdTom; Pvalb:Cre

Wrote 30 384-well entries

Successfully wrote new 384-well plate manifest to: ../Downloads/retype_manifest_384.csv

Clear session

Upload matching Stage3.csv file here

Select the matching Stage3.csv file

Drag and drop file here

Limit 200MB per file

Browse files

Stage3.csv

153.7KB

X

The manifest is automatically saved in the Downloads folder as "retype_manifest_384.csv"

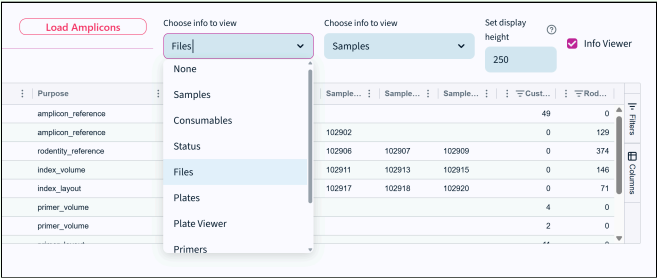
Make sure you place this somewhere safe before you use Retype again, because it will overwrite this file without

You should upload it to a new experiment in NGSG Explorer using the Stage 1: Load, Custom 384 well manifest option.



NGSG Explorer Information Viewer Panels

Throughout the NGSG Explorer interface, there are information panels that can be used to display various types of information, and provide controls for selecting and removing items from the experiment.



These panels are controlled by drop-down selectors, and can be swapped out as needed by the users. There is a control provided to shrink or grow the height of panels for convenience. Columns within a panel can be adjusted to show or hide specific information.

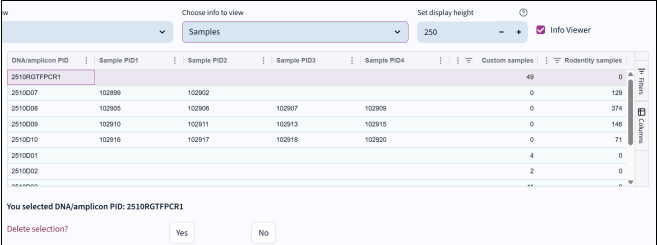
Specific Information Panels

- [None](#)
- [Samples](#)
- [Consumables](#)
- [Status](#)
- [Files](#)
- [Plates](#)
- [Plate Viewer](#)
- [Primers](#)
- [Indexes](#)
- [Log](#)

None

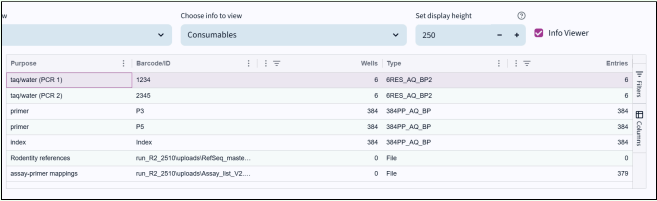
This disables the information panel, allowing the full horizontal space to be used by another information panel. Simply changing the selection to another panel type will reactivate the panel.

Samples



This panel displays all samples that have been loaded into the experiment, along with their relevant information such as sample name, sample ID, and any associated metadata. This panel allows users to select samples to remove them from the experiment.

Consumables



This panel displays all consumables that have been loaded into the experiment, such as primers, sequencing indexes, and information files, required for the genotyping process. Users can view details about each consumable, such as id, filename, relevant quantity.

Status

Choose info to view			Set display height		Info Viewer
Status			250		
Stage order	Staged file	Affected plates	Status		
1	run_R2_2306(stage1_P2306054.csv	230123VRP1, 230123VRP2, 230123VRP3, 230...	committed		
1	run_R2_2306(indexLayout_230923_GTF.fa	230123VRP5, 230123VRP6	committed		
1	run_R2_2306(stage1_P2306050.csv	230123VRP5, 230123VRP6	committed		
2	run_R2_2306(pending_stage1.csv		committed		
2	run_R2_2306(pending_PCR2_index-picked_R2	Index	committed		
2	run_R2_2306(pending_PCR2_index-pickd_1234	1234	committed		
2	run_R2_2306(pending_Midway_R2_2308.csv		committed		

This panel provides an overview of the current status of the experiment, notably which pipeline sections have been run. It allows users to quickly assess the progress of the experiment so that they can carry on from the last completed stage.

Files

Choose info to view				Set display height		Info Viewer
Files				250		
File	Date modified	Purpose				
run_R2_2510(uploadRefSeq_masterV3_250923_GTF_171025.fa	2025/10/17 14:51:16	amplicon_reference				
run_R2_2510(uploadRefSeq_masterV3_250923_GTF.fa	2025/10/16 12:50:51	amplicon_reference				
run_R2_2510(uploadRefSeq_masterV3_251008.fa	2025/10/16 12:19:03	identity_reference				
run_R2_2510(uploadIndex_plateSurvey_fuid_volume_176PLAT...	2025/10/13 10:20:54	index_volume				
run_R2_2510(uploadIndex_plateSurvey_fuid_volume_176PLAT...	2025/10/13 10:20:54	index_layout				
run_R2_2510(uploadIndex_plateSurvey_fuid_volume_20200226.cav	2025/10/13 10:20:10	primer_volume				
run_R2_2510(uploadIndex_plateSurvey_fuid_volume_20200226.cav	2025/10/13 10:20:10	primer_volume				

You selected ["run_R2_2510/uploads/RefSeq_masterV3_250923_GTF.fa"]

Delete selection?

Yes
 No

This panel displays all files that have been uploaded or generated during the experiment, such as raw data files, intermediate results, and final reports. Users can view details about each file, such as name, type, and size, and can select files to remove them from the experiment if they are no longer needed.

Plates

Choose info to view				Set display height		Info Viewer
Plates				250		
Plate	Num Wells	Purpose				
102899	92	sample				
102902	37	sample				
2510D07	129	dna				
102905	93	sample				
102906	93	sample				
102907	93	sample				
102909	93	sample				

You selected ["102899"]

Delete selection?

Yes
 No

This panel displays all plates that have been loaded into the experiment, along with their relevant information such as plate name, plate type, and the samples or reagents contained within each plate. Users can select plates to remove them from the experiment if they are no longer needed.

Plate Viewer

This panel provides a visual representation of the plates used in the experiment. You can select any loaded plate to view its layout and contents.

Choose info to view

Plate Viewer

Set display height

250

+

-

Info Viewer

Plate ID to view

sample plate: 102899

sample plate: 102899

sample plate: 102902

dna plate: 2510D07

sample plate: 102905

sample plate: 102906

sample plate: 102907

sample plate: 102909

dna plate: 2510D08

It will then display the plate layout.

Plate ID to view

sample plate: 102899

	1	3	5	7	9	11						
A	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
B	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12
C	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
D	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12
E	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	E11	E12
F	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12
G	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12
H	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12

<

Users can mouse-over each well to see the layout of samples and reagents within each plate. It can be used to verify that samples and reagents are correctly arranged and to identify any potential issues with plate setup.

Primers

Choose info to view: Primers Set display height: 250 Info Viewer

Primer	Required Dos...	Required Vol...	Required Wells	Available Dos...	Available Vol...	Available Wells	Positions
Pipet	35	8.75	1	480	120	4	P5, O16L10,N10,M18
U4-act	40	10	1	480	120	4	P5, O24,P24,O23,O23
MHCII_WT	6	1.5	1	120	30	1	P5, I5
MHCII_MUT	6	1.5	1	120	30	1	P5, J5
Tx1-FQMB	26	6.5	1	120	30	1	P5, P18
Mm2	0	0	0	120	30	1	P5, D23
Cep-dna1-4x5-FQMB	0	0	0	120	30	1	P5, E3

Primer table written to file: run_R2_2510primer_list.csv

This panel displays all primers that have been loaded into the experiment, along with their relevant information such as primer name, sequence, and any associated metadata. The user cannot remove these primer wells once they have been loaded. The plate must be overwritten with one of the same name, or the old one removed and a new plate layout and volume file must be uploaded.

Indexes

This panel displays all indexes that have been loaded into the experiment, along with their relevant information such as index name, sequence, and any associated metadata. The user cannot remove these index wells once they have been loaded. The plate must be overwritten with one of the same name, or the old one removed and a new plate layout and volume file must be uploaded.

Activity Log

This panel displays the activity log for the experiment, providing a record of all actions taken within the experiment, along with any warnings or errors that may have occurred. Users can use this panel to track the progress of the experiment and identify any issues that may need to be addressed.

At the bottom of every page is another information panel that always defaults to displaying the activity log.

Choose info to view: Log Choose info to view: None Set display height: 350

Time	Level	Message	Function name	Func line
2025-11-20 12:41	Success	created primer assay families file run_R2_2510primers.csv	generate_primer_assayfams	1181
2025-11-20 12:41	Success	Created reference sequences file run_R2_2510targets.fa containing 882 sequences	generate_targets	1149
2025-11-04 15:11	Info	C:\Users\Cami\NGSgenovenv\ngsenv\Scripts\python.exe bin\ngsmatch.py --ncpus 8 --rundir run_R2_2510 --margin 0.9 --ide...	main	1104
2025-11-04 15:11	Success	created primer assay families file run_R2_2510primers.csv	generate_primer_assayfams	1181
2025-11-04 15:11	Success	Created reference sequences file run_R2_2510targets.fa containing 882 sequences	generate_targets	1149
2025-10-30 13:32	Info	C:\Users\Cami\NGSgenovenv\ngsenv\Scripts\python.exe bin\ngsmatch.py --ncpus 8 --rundir run_R2_2510 --margin 0.9 --ide...	main	1106
2025-10-30 13:32	Success	created primer assay families file run_R2_2510primers.csv	generate_primer_assayfams	1181
2025-10-30 13:32	Success	Created reference sequences file run_R2_2510targets.fa containing 882 sequences	generate_targets	1149
2025-10-30 13:27	Info	C:\Users\Cami\NGSgenovenv\ngsenv\Scripts\python.exe bin\ngsmatch.py --ncpus 8 --rundir run_R2_2510 --margin 0.9 --ide...	main	1106
2025-10-30 13:27	Success	created primer assay families file run_R2_2510primers.csv	generate_primer_assayfams	1181
2025-10-30 13:27	Success	Created reference sequences file run_R2_2510targets.fa containing 882 sequences	generate_targets	1149

This is a crucial tool for tracking all actions taken within the experiment. It is possible to change the view from the log panel to another information panel type, and you can have a second panel visible too. By default only the log is shown here to provide sufficient horizontal screen space.

The columns in the log display are (from left to right): Time, Level, Message, Function name, Func line.

- Time - the time and date the log entry was created
- Level - the severity level of the log entry (e.g. info, warning, error)
- Message - a descriptive message about the log entry
- Function name - the name of the function where the log entry was generated
- Func line - the programming line number in the function where the log entry was generated

Message levels are used to indicate the severity of an issue.

- Info - informational messages providing general updates on events
- Success - indicates successful completion of an action or stage
- Failure - indicates a failed action or stage. These are not typically problematic
- Warning - indicates a potential issue that may not stop the pipeline from running, such as insufficient primer volumes, but should be considered
- Error - indicates an issue that will prevent the pipeline from running, such as missing files or incorrect file formats
- Critical - indicates a severe issue that requires immediate attention, such as system failures or data corruption. These indicate that the pipeline is in an inconsistent state or has encountered a programming error. You should raise any critical issues immediately.
- Debug - detailed information for debugging purposes - unlikely to be seen in regular use

To aid users in spotting important issues, **warnings** are displayed in yellow, **errors** are displayed in red, and **critical** messages are in purple.

You cannot remove or clear log entries.



Input and output file formats

This page provides detailed information about the input and output file formats used in the NGS Genotyping Pipeline.

User Files

Users will need to upload various files at different stages of the pipeline, and also download files generated by the pipeline to USB for use with the Hamilton Nimbus robot and Illumina Miseq sequencer.

These files often have specific structures and name requirements that must be followed for the pipeline to process them correctly, or at least easy for users to recognise. Take the time to clearly name files for their intended purpose.

Some critical terms are used in the file names (and are noted in the specific entries below). If a user accidentally tries to upload a file for a different purpose e.g. primer layout file being uploaded as a primer volume file, the pipeline will reject it with a warning, and suggest the correct file type.

Always separate filename components with underscores. Never use spaces in filenames, use a dash '-' instead if necessary.

If a file of the same name is uploaded, the pipeline will ask the user whether they want to overwrite the existing file.

The term "manifest" is used in this pipeline to refer to input files that may contain records for more than one plate of samples

- [Rodentity Ear Punch Plate files](#)
- [Custom Sample 96-Well plate manifest](#)
- [Custom Sample 384-well plate manifest](#)
- [384-well Amplicon manifest](#)
- [Nimbus input files](#)
- [Echo 384 COC files](#)
- [Primer Layout input files](#)
- [Primer volume input files](#)
- [Primer survey file](#)
- [Primer list file](#)
- [Index layout input files](#)
- [Index volume input files](#)
- [Index survey file](#)
- [Miseq sample sheet file](#)
- [FASTQ sequencing files](#)
- [Reference genome files \(FASTA\)](#)
- [Assay List files](#)
- [Variant Sequences file](#)
- [Results Workbook \(CSV\) aka Genotyping Report](#)

Internal Files

The pipeline generates various internal files during its operation, which are used for tracking and managing the genotyping process.

- [Stage1 CSV](#)
- [Stage2 CSV](#)
- [Stage3 CSV](#)
- [Results CSV](#)
- [Amplicon Results CSV](#)
- [Reported well text files](#)
- [debug.log](#)
- [Targets CSV](#)
- [Amplicon Targets CSV](#)
- [Matching Log](#)
- [Filter Parameters TSV files](#)
- [Experiment JSON](#)

Rodentity Ear Punch Plates 96-well

These files are JSON format and are downloaded from the Rodentity database (DB), describing the 96-well plates used for mouse ear punch samples. They are a complex file format that is provided directly by Rodentity and should not be edited by users. They contain information about the plate layout, sample IDs, and other metadata.

These files are uploaded in the Load Samples tab, in the first stage of the pipeline.

Here is an example of the structure of these files, which must be parsed by the pipeline code on loading. Actual names have been replaced with descriptive placeholders.

```
{
  "wells": [
    {
      "plate": "NNNNNN INT",
      "wellLocation": "A1",
      "mouse": {
        "mouseKey": INT,
        "mouseLineName": "Text",
        "barcode": "M and long number as a string",
        "mouseNumber": INT,
        "sex": "M",
        "generation": "N1",
        "dateOfBirth": "2024-12-02T00:00:00.000Z",
        "mouseState": "word",
        "alleles": [
```

```
{
  "alleleKey": INT,
  "name": "Long text",
  "symbol": "Short text",
  "options": [
    Standard genotype options as strings, e.g. "m/+", "m/m"
  ],
  "assays": [
    {
      "assay_key": INT,
      "name": "text",
      "method": "string type of assay e.g. NGS"
    },
  ]
},
```

[Back to Top](#)

Custom Sample 96-well Plate Manifest

These are comma-separated CSV files that users can create themselves to load custom sample plates that are not defined by Rodentity.

They have the same format as the Custom Sample 384-well format, only accept 96-well plates.

They are uploaded in the Load Samples tab, between the Rodentity and Custom Sample 384-well plate manifest sections, in the first stage of the pipeline.

Required columns: [plateBarcode, well, sampleBarcode, assay*, clientName] *Duplicates allowed Optional columns: [sampleNo, sampleName, alleleSymbol] and anything else

Example headers:

sampleNum	plateBarcode	well	sampleBarcode	assay	assay	clientName	sampleName	alleleSymbol
-----------	--------------	------	---------------	-------	-------	------------	------------	--------------

[Back to Top](#)

Custom Sample 384-well Plate Manifest

These are comma-separated CSV files generated by NGSG Retype, describing a set of 384-well plates of DNA samples that are to be resequenced, though they could also be defined manually by users.

They have the same basic format as the Custom Sample 96-well format, but only accept 384-well plates.

Required columns: [plateBarcode, well, sampleBarcode, assay*, clientName] *Duplicates allowed Optional columns: [sampleNo, sampleName, alleleSymbol] and anything else

Example headers:

sampleNum	plateBarcode	well	sampleBarcode	assay	assay	clientName	sampleName	alleleSymbol
-----------	--------------	------	---------------	-------	-------	------------	------------	--------------

[Back to Top](#)

384-well Amplicon Manifest

This is a CSV file format that describes the amplicons loaded in the 384-well amplicon plates.

The amplicon columns refer to assays defined in the Amplicon reference list file. You can upload these in the Load Amplicons tab of the Load stage.

There is no primer amplification stage needed for amplicon samples - they skip straight to indexing and sequencing.

Example file:

```
sampleNo,plateBarcode,well,sampleBarcode,amplicon,amplicon,amplicon,amplicon,sample name
1,2510RGTFPCR1,A1,C250716LM01,Brachy-K0,,,,256-1
```

[Back to Top](#)

Nimbus input files

These are comma-separated CSV files generated by the pipeline and downloaded by the user to USB.

Nimbus files are named "Nimbus-plateID.csv", where plateID is the actual 384-well plate identifier that the Nimbus will aggregate samples into.

They inform the Hamilton Nimbus robot about the samples in each plate. The robot decides which position in the output plate to place each sample. The robot will then write an Echo_384_COC file which describes the transfer movements.

```
"Sample no","Plate barcode","Well","Sample barcode"
"1","102899","A1","M043782100257"
```

[Back to Top](#)

Echo 384 COC Files

This is a CSV file format that indirectly describes the DNA samples loaded in the 384-well DNA plates. In order to generate a record of the 384-well Echo plate, the movements conducted by the Hamilton Nimbus robot are recorded in the Echo_384_COC file (CSV file format) that describes the source and destination of each transfer, while the sample and assay information is taken from the Rodentity or custom manifest records describing the 96-well sample plates. Descriptions of these are generated by the Hamilton Nimbus robot when it combines input from 1-4 96-well sample plates.

These are also used as inputs for resequencing failed samples/assays (using the NGSG Retype workflow).

```
"RecordId","TRackBC","TLabwareId","TPositionId","SRackBC","SLabwareId","SPositionId","SPositionBC"
1,"2510D07","Echo_384_COC_0001","A1","102899","ABg_96_PCR_NoSkirt_0001","A1","M043782100257M"
```

[Back to Top](#)

Primer Layout input files

These are comma-separated CSV file formats that describe the layout of primers in the 384-well primer plates.

The first part of the file name is considered as the plate barcode or identifier. The words "primer" and "layout" should be included in the file name. All parts of the file name must be separated by underscores. e.g P3_Primer_layout_384-well_Echo_pool_20250718.csv

```
plate position on Echo 384 PP,primer names pooled
A01,Casp1-10del
```

The header column names don't matter and are ignored by the parsing code.

[Back to Top](#)

Primer volume input files

These are comma-separated CSV file formats that describe the volumes of primers available to be dispensed into the Echo PCR plates.

The first part of the file name is the plate identifier or barcode, and must match the plate identifier in the primer layout file.

All parts of the file name must be separated by underscores. e.g P3_Primer_volumes_384-well_Echo_pool_20250718.csv

```
Date,3/12/2020,,,,,,,,,,,,,
Time,6:38:17,,,,,,,,,,,,,
,,,,,,,,,,,,,
,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24
A,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
B,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
C,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
D,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
E,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
F,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
G,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
H,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
I,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
J,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
K,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
L,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
M,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
N,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
O,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
P,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60,60
```

Alternatively, the volume layout can be provided as a comma-separated CSV file with two columns: "well" and "volume".

[Back to Top](#)

Primer survey file

This is a comma-separated CSV file that describes the primers available for use in the genotyping process. Having this as a list can be helpful for the lab team in setting up primer plates.

It is named "Primer-svy.csv". Here is the format:

```
Source Plate Name,Source Plate Barcode,Source Plate Type,plate position on Echo 384 PP,primer names pooled,volume
Source[1],P3,384PP_AQ_BP,A1,Casp1-10del,60.0
```

[Back to Top](#)

Primer list file

This is a comma-separated CSV file that describes the primers available for use in the genotyping process, but in a much more detailed format than the primer survey file. It includes information about the required and available volumes and doses of each primer, as well as the positions of the primers in the Primer plates that have been uploaded to the pipeline so far.

It is named "Primer-list.csv". Here is the format:

```
Primer,Required Doses,Required Volume (µL),Required Wells,Available Doses,Available Volume (µL),Available Wells,Positions,::auto_unique_id::
Ptprc,35,8.75,1,480.0,120.0,4,"P3:018,L18,N18,M18",0
```

[Back to Top](#)

Index layout input files

These are comma-separated CSV file formats that describe the layout of index barcodes in the 384-well index plates.

```
Source well,Name for IDT ordering,index ,Name,Oligo * -phosphothioate modif. against exonuclease,
A1,96_SET_1_i7F_1,AACCTTGG,index_8nt_7,CAAGCAGAAGACGGCATACGAGATccaaggttGTCTCGTGGGCTCG*G,
```

[Back to Top](#)

Index volume input files

These are CSV file formats that describe the volumes of index barcodes available to be dispensed into the Echo PCR plates. They are separate from the layouts as they can be created from measurements taken by the Echo directly.

In practice, a standard pair of layout and volume files is used for all assays, without measurement.

```
Date,3/12/2020,,,,,,,,,,,,,
Time,8:50:31,,,,,,,,,,,,,
,,,,,,,,,,,,,
,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24
A,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,0,0,0,0
B,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,0,0,0,0
C,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,0,0,0,0
D,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,56,0,0,0,0
E,56,0,56,0,56,0,56,0,56,56,56,56,0,56,0,56,0,56,56,56,56,0,0,0,0
F,56,0,56,0,56,0,56,56,56,56,0,56,0,56,0,56,56,56,56,56,56,0,0,0,0
G,56,0,56,0,56,0,56,56,56,56,0,56,0,56,0,56,56,56,56,56,56,0,0,0,0
H,56,0,56,0,56,0,56,56,56,56,0,56,0,56,0,56,56,56,56,56,56,0,0,0,0
I,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
J,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
K,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
L,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
M,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
N,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
O,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
P,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
```

Alternatively, the volume layout can be provided as a comma-separated CSV file with two columns: "well" and "volume".

[Back to Top](#)

Index survey file

This is a comma-separated CSV file that describes the index barcodes available for use in the genotyping process. It is generated automatically by the pipeline to help with lab preparation for the indexing process.

It is named "Index-svy.csv". Here is the format:

```
Source Plate Name,Source Plate Barcode,Source Plate Type,Source well,Name for IDT ordering,index,Name,Oligo * -phosphothioate modif. against exonuclease,v
Source[1],Index,384PP_AQ_BP,A1,96_SET_1_i7F_1,AACCTTGG,index_8nt_7,CAAGCAGAAGACGGCATACGAGATccaaggttGTCTCGTGGGCTCG*G,56.0
```

[Back to Top](#)

Miseq sample sheet file

This is a comma-separated CSV file that describes the samples and index barcodes loaded in the Miseq sequencer. It is generated by the pipeline and downloaded by the user to USB, and then uploaded back to the pipeline after sequencing to inform the pipeline about the sample layout in the sequencer.

It is named "Miseq_runID.csv", where "runID" is the unique identifier/experiment name within NGSG Explorer.

Here is an example file with a single sample entry:

```
[Header],,,,,,,,,,
IEMFileVersion,4,,,,,,,,,
Investigator Name,,,,,,,,,
Experiment Name,NGSG:R2_2510,,,,,,,,,
Date,,,,,,,,,
Workflow,GenerateFASTQ,,,,,,,,,
Application,FASTQ Only,,,,,,,,,
Assay,Nextera XT,,,,,,,,,
Description,Mouse genotyping,,,,,,,,,
Chemistry,Amplicon,,,,,,,,,
,,,,,,,,,
[Reads],,,,,,,,,,
151,,,,,,,,,
151,,,,,,,,,
,,,,,,,,,
[Settings],,,,,,,,,,
ReverseComplement,0,,,,,,,,,
Adapter,,,,,,,,,
,,,,,,,,,
[Data],,,,,,,,,,
Sample_ID,Sample_Name,Sample_Plate,Sample_Well,I7_Index_ID,index,I5_Index_ID,index2,Sample_Project,Description
2510R2PCR1_A01,M043773700032,,,96_SET_1_i7F_1,AACCTTGG,96_SET_1_i5R_1,AACCGAAC,NGSgeno
```

[Back to Top](#)

FASTQ sequencing files

These are standard FASTQ files generated by the Illumina Miseq platform, containing the sequenced DNA reads from the samples. They are used as input for the sequence matching stage of the pipeline.

These may have file extensions of .fastq or .fq, and may also be compressed with gzip (.gz), e.g. .fastq.gz or .fq.gz

[Back to Top](#)

Reference genome files (FASTA)

These are standard FASTA files containing reference DNA sequences for the amplicons being analyzed. They are used as an input for the sequence matching stage of the pipeline to compare against the sequenced reads and determine genotypes.

These files may have file extensions of .fasta or .fa, and may also be compressed with gzip (.gz), e.g. .fasta.gz or .fa.gz

As of NGS Genotyping Pipeline version 2, multiple reference genome files can be uploaded.. Separate reference files are typically used for Rodentity and Amplicon plate matching.

The NGS Genotyping Pipeline incorporates an extension to the FASTA format by allowing '(', '[', and '],')' characters in the sequence definitions. These denote repeat regions that may be present in zero or more copies and are not required in matched sequences. For example:

```
>ABC1_target-with-repeat
CGAGACATTCGAGTGTGCGCGCAGGCGAGAGTTTTGAGAGCGCGAGCTTTCAGGAGACAGGCGAACGCAGCTTACGTAGCAGCT(C)CGAAGTTTCGAGCAGAGC
```

[Back to Top](#)

Assay List files

These are comma-separated CSV files that describe the relationship between mouse NGS assays, the family of assays that must be performed to correctly identity the family of alleles, and the primer names that must be loaded that specific assay.

Here is an example with a single entry. Any number of primer columns can be included. There must be at least three columns populated in each row.

assay	assayFamily	primer	primer	primer	primer	primer	primer	primer
Ptprc	Ptprc	Ptprc						

[Back to Top](#)

Variant Sequences file (FASTA)

This is a FASTA file that enumerates all unexpected sequences for a given sample number.

If the sequence doesn't match any expected target, it will give the full sequence (and the count in the header). If it matches a known but unexpected sequence, which may be a contaminant, then the name is given instead of the sequence.

```
>Sample:2;count:76
TCCCATTTTATCACTGACTTCCTGCTTCTCTCTCCCTCGACACGCAGAGCCAACCCACTGATTATTACGTTCTGGGAGAAGAGGAGGAGGAGGAAGAGGAAGAGGTGTGTGGGGATGAGGATGAGGCTGGCC
>Sample:2;count:76
GCTCCCATTTTATCACTGACTTCCTGCTTCTCTCTCCCTCGACACGCAGAGCCAACCCACTGATTATTACGTTCTGGGAGAAGAGGAGGGGAGGAAGAGGAAGAGGTGTGTGGGGATGAGGATGAGGCTGGCC
>Sample:2;count:76
Tap1Tap2-K0_MUT-K0:K0
```

Results workbook (aka Rodentity Genotype report)

This is a comma-separated CSV file that is returned by a Rodentity Genotyping Web Service.

It can be fed back into the NGSG Retype web tool, along with the Stage3.csv file from the matching experiment run folder, to automatically pick out failed sample-primer combinations for re-typing.

The format is not defined within this code base, but is instead the property of the Rodentity support team (APF).

```
barcode,code_assays,plate,wellLocation,sex,alleleSymbol,alleleKey,assayKey,passFail,seqName1,seqName2,seqName3,seqName4,seqName5,seqName6,seqName7,efficien
M035658200149,M035658200149;R26-TdTomato,96990,B8,F,Gt(R0SA)26Sor:TdTom,386,2158,pass,,,,,,,,,0,0,0,?,Y,
```

Internal files generated by the pipeline

These are files generated by the pipeline during its operation, which are used for tracking and managing the genotyping process. Some of these files are required for the pipeline to function, while others are generated for debugging or record-keeping purposes.

Users are unlikely to need to interact with these files directly, but their details are included here for pipeline maintainers.

Stage1 CSV

These are comma-separated CSV files generated by the pipeline to track the samples and assays loaded in stage 1 of the pipeline and are generated in 1-1 correspondance with Nimbus-*plateID*.csv files. The used to be the staging files for the actual pipeline, were replaced with an experiment JSON file to allow for more flexible experiment design. They are no longer strictly required by the pipeline and may be removed from a future release, though they do form the basis for later Stage2.csv and Stage3.csv files.

Example file, note the "p" guarded plate barcodes and the "r" sample barcodes:

```
"samplePlate","sampleWell","sampleBarcode","strain","sex","alleleSymbol","alleleKey","assayKey","assays","assayFamilies","clientName","sampleName"
"p102899p","C2","rM043732400027r","OT-I:CD45.1","M","""Ptprc:ab""","1559","2020","Ptprc","Ptprc","",""
```

[Back to Top](#)

Stage2 CSV

This is a comma-separated CSV file generated by the pipeline to track the samples after the primer-allocation stage (PCR1) of the pipeline. There is only a single Stage2.csv file and it extends the Stage1-plateID.csv file with the following columns:

- dnaPlate - the ID of the 384-well plate containing the DNA samples (from Nimbus)
- dnaWell - the well location of the DNA sample in the 384-well plate (from Nimbus)
- primer - the primer name to be dispensed onto the Echo PCR plate
- primerPlate - the plate on which the primer is located
- primerWell - the well on the primer plate
- pcrPlate - the (Echo 384) plate on which the primer PCR reaction is performed
- pcrWell - the well on the PCR plate

The Stage2.csv file is currently required by the pipeline for the indexing stage. This requirement should be removed in future versions, resulting this file being deprecated.

Example file (just the last seven columns):

```
"dnaPlate","dnaWell","primer","primerPlate","primerWell","pcrPlate","pcrWell"
"p2510D07p","A2","Ubt-f-ex8-F0M2","pP5p","C7","p2510R2PCR1p","A1"
```

[Back to Top](#)

Stage3 CSV

This is a comma-separated CSV file generated by the pipeline to track the samples after index allocation (PCR2). There is only a single Stage3.csv file and it extends the Stage2.csv file with the following columns:

- i7bc - the i7 barcode sequence
- i7name - the name of the i7 barcode
- i7well - the well on the i7 barcode plate
- i5bc - the i5 barcode sequence
- i5name - the name of the i5 barcode
- i5well - the well on the i5 barcode plate
- index_plate - the plate on which the index is located

Stage3.csv is currently required by the pipeline and forms the basis for the final results.csv and amplicon_results.csv files.

Example file: just the last seven columns

```
"i7bc","i7name","i7well","i5bc","i5name","i5well","index_plate"
"AACCTTGG","96_SET_1_i7F_1","A1","AACCGAAC","96_SET_1_i5R_1","A7","Index"
```

[Back to Top](#)

Results CSV

This is a comma-separated CSV file generated by the pipeline to report the genotyping results for all sequenced wells of Echo PCR plates used in an experiment run.

Note: there is no separation of Rodentity and Custom results as of v2.04.000

The results.csv file extends Stage3.CSV with the following columns:

- readCount: the number of reads obtained for the sample
- cleanCount: the number of reads obtained after cleaning low-quality bases and very short reads
- mergeCount: the number of read-ends that were successfully merged for the sample
- seqCount: the number of merged reads assigned to a paired seqName in the next column, ';' separated
- seqName: the name of the sequenced read paired to a given count in the previous column, ';' separated
- efficiency: the efficiency of the genotyping process = the proportion of reads that matched an expected genotype
- otherCount: the number of reads obtained that matched to an alternative known sequence, or an unknown sequence
- otherName: the name of the alternative or unknown sequence. If significant, the actual sequence will be listed. Insignificant matches will be indicated as "other" and the number of different sequences displayed in parentheses

Example file, just the last eight columns:

```
"readCount","cleanCount","mergeCount","seqCount","seqName","efficiency","otherCount","otherName"
"19911","19568","19241","6560;4106","UbtF-ex8-F0M2_MUT-del20:;UbtF-ex8-F0M2_WT","0.554","8575","other (2397)"
```

[Back to Top](#)

Reported Well Text Files

These are text files containing information about wells that have been reported (and written to PDF) in the experiment.

They are available for amplicons, Rodentity, and custom samples, and take the form amplicon_reported.txt, etc.

They ensure that we don't double-up entries that have been multiple sequence aligned and then written to a PDF report.

Here is an example entry:

```
2510RGTFPCR1      E3      C250929CM73901  Stag2-K0
```

[Back to Top](#)

Amplicon Results CSV

This is identical to the Results CSV file, but only includes amplicon samples. Information regarding DNA plates or primer plate details are likely to be missing.

[Back to Top](#)

debug.log

This is a log file generated by the Allele matching process, when enabled by the user.

Each entry is led by the log level and a colon. Usually the process ID follows, and then whatever the programmer deems useful.

Here is an actual example line from this file:

```
Info: 15916 Working on sampleNo='1' pcrPlate='2510R2PCR1' pcrWell='A01'
```

[Back to Top](#)

Target Sequences file (FASTA)

A subset of all the available reference sequence files that have been loaded, to be used for matching the Rodentity and Custom samples. It follows the exact same format as the reference sequence files.

Amplicon Target Sequences file (FASTA)

A subset of all the available reference sequence files that have been loaded, to be used for matching the amplicon samples. It follows the exact same format as the reference sequence files.

Matching Log

This is the default logging of the allele matching process. It otherwise resembles the debug.log file.

[Back to Top](#)

Filter Parameters TSV files

These are tab-separated value files containing the parameters used for filtering the view of samples, primarily for subsequent multiple sequence alignment (disabled as of v2.04.000). They are generated automatically during the reporting stage as the user adjusts filtering criteria.

Separate files exist for each sample group:

- Rodentity samples - rodentity_table_filter_params.txt
- Custom samples - custom_table_filter_params.txt
- Amplicon samples - amplicon_table_filter_params.txt

All have the same basic format of plateID, wellID, sampleID, and proportion count cutoff:

102899 B4 M043738200004 0.15

[Back to Top](#)

Experiment JSON file

This is a JSON file containing the experimental design and metadata for the entire experiment.

While the empty experiment file is a minimal representation of the experiment structure, it rapidly gets very complex with nested dictionary structures as data is loaded. Understanding a fully populated experiment file requires familiarity with the internal workings of the pipeline, and is not necessary for users of the pipeline.

```
{
  "py/object": "experiment.Experiment",
  "py/state": {
    "name": "test1",
    "description": "",
    "locked": false,
    "unassigned_plates": {
      "custom": {},
      "json://1": "",
      "json://2": "",
      "json://3": "",
      "json://4": ""
    },
    "dest_sample_plates": {},
    "plate_location_sample": {},
    "deleted_plates": {},
    "stage": "new",
    "denied_assays": [],
    "denied_primers": [],
    "assay_synonyms": {},
    "primer_assayfam": {},
    "assayfam_primers": {},
    "assay_assayfam": {},
    "assayfam_assays": {},
    "reference_sequences": {},
    "uploaded_files": {
      "_upload_queue": {},
      "_upload_pending": {}
    },
    "reproducible_steps": [],
    "pending_steps": {
      "py/set": []
    }
  }
}
```

```
},
"pending_uploads": {
  "py/set": []
},
"log_entries": [],
"log_time": {
  "py/object": "datetime.datetime",
  "__reduce__": [
    {
      "py/type": "datetime.datetime"
    },
    [
      "B+oEDg0FAwSFSw=="
    ]
  ]
},
"transfer_volumes": {
  "DNA_VOL": 200,
  "PRIMER_VOL": 250,
  "PRIMER_TAQ_VOL": 1000,
  "PRIMER_WATER_VOL": 550,
  "INDEX_VOL": 175,
  "INDEX_TAQ_VOL": 2000,
  "INDEX_WATER_VOL": 650
}
}
```

[Back to Top](#)



The NGS Explorer Sequence Matching

Sequence matching is a critical step in the NGS genotyping pipeline that involves comparing the sequencing reads against a reference database to identify the presence or absence of specific genetic variants.

In the NGS Genotyping pipeline, sequence matching is done with a process of several steps. These are launched in parallel, across several physical computer processors, to save time. Steps that involve long wait times are also performed asynchronously where possible. This means that code execution "threads" can continue to work on other tasks while waiting for the completion of a long-running task, such as a sequence quality trimming or read-end merging. This allows for more efficient use of computational resources and can significantly reduce the overall time required for the sequence matching process.

Repeating what is written in the Alleles Stage section... When the "Run allele calling" button is pressed, the following steps are performed:

1. All paired-end FASTQ sequence record (two files per well) are trimmed of low quality bases, and short reads are filtered out
2. Clean paired sequences are merged. Unmerged reads are saved, but not included in later processing
3. Merged sequences are matched against the reference sequences expected for them given the primers that were used
4. If a particular merged sequence wasn't seen at least 5 times, or has less 10% of the most common expected sequence (using the default parameters) then it is skipped to save time
5. By default sequences must exactly match the expected reference sequence. They can be a little shorter, or longer, but the overlapping majority must be identical
6. A results.csv file is created that contains all the wells that were expected and had a matching pair of FASTQ sequence files

Steps 1 and 2 are performed using BBtools, by Brian Bushnell. The tools bbdduk and bbmerge are used for quality trimming and read merging, respectively.

The bbdduk command parameters are: -Xmx1g t=2 qtrim=rl trimq=20 minlen=50 k=23 ktrim=r mink=11 hdist=1 overwrite=true ref=bbmap\resources\adapters.fa

These specify:

1. -Xmx1g: Use 1 GB of memory
2. t=2: Use 2 threads
3. qtrim=rl: Trim both ends of the reads based on quality
4. trimq=20: Trim bases with quality below 20
5. minlen=50: Discard reads shorter than 50 bases after trimming
6. k=23: Use a k-mer size of 23 for adapter trimming
7. ktrim=r: Trim adapters from the right end of the reads
8. mink=11: Use a minimum k-mer size of 11 for adapter trimming
9. hdist=1: Allow for 1 mismatch when matching adapter k-mers
10. overwrite=true: Overwrite existing output files
11. ref=bbmap\resources\adapters.fa: Use the specified adapter reference file for trimming

bbmerge is only run with one non-standard parameter: -pfilter=1, which forces perfect overlap requirements. This ensures that no artifical sequences are generated from the merging process.

Step 3 is matching. In the main genotyping pipeline, this is always done using **exact** matching. Inexact matching is optional, and is always used in the amplicon pipeline. These are described in more detail below.

Step 4 is a time-saving filter. Reads that do not meet the minimum criteria are skipped to save computational resources. Where to set the minimum read coverage, or the minimum proportion of reads will be determined by the PCR reaction likelihood and must be determined by trial and error. In the future, it would be ideal to set these parameters based on past performance of individual primers, but at present this kind of capability isn't available.

Exact matching

Exact matching means that if seq A matches somewhere within the length of seq B (or vice versa) then this is a match. No differences are allowed within the overlapping region. This may seem strict, but around 10%-40% of merged reads will match their reference sequence this way. Exact matching is very fast, and is always performed as a preliminary step before inexact matching.

By default a maximum length difference of 10% is allowed between the merged read and the reference sequence. That is to say, the shorter sequence must be at least 90% the length of the longer sequence.

Inexact matching

As an optional feature for the genotyping pipeline, inexact matching can be used to allow for some differences between the merged sequences and the reference sequences.

Inexact matching is always done in two round, firstly against any expected sequences, and then against all remaining reference sequences.

This is partly done to improve performance, but also to avoid false positive detection against similar but unrelated sequences.

Inexact matching uses the Bio.Align module from the Biopython library v1.85 to perform global (end-to-end) sequence alignment using the following parameters:

- aligner.mode = 'global'
- aligner.open_gap_score = -2.0
- aligner.extend_gap_score = -0.7
- aligner.end_gap_score = -0.5
- aligner.match_score = 3.5
- aligner.mismatch_score = -1.0

These parameters allow for a more flexible matching approach, accommodating sequencing errors and polymorphisms, which are then annotated after the best match is found.

In order to be declared a match, the alignment score must be 90% (by default) of the maximum possible score. This cutoff can be set in the interface before the matching is started.

Local alignment should not be used for amplicons in general, as it is likely to produce false matches against unrelated sequences.

The settings above are based on defaults values used in the BWA aligner by Heng Li, and are otherwise chosen empirically to fit with Illumina sequencing data.

Inexact matching is always turned on with the amplicon pipeline, as we always want to find and report variants.

In any case, exact matching is always performed as a preliminary step as it's much faster, followed by inexact matching if necessary.